Jess Sullivan

Full Stack Engineer · DevSecOps · Computer Vision · Kernel & Security Research · ML/HPC

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github.com/jesssullivan (74+ stars, long-standing FOSS contributor) · transscendsurvival.org · LinkedIn

Senior full-stack and systems engineer with 8+ years building software across TypeScript/SvelteKit, Python, Go, C++, Rust, Zig, Chapel, Haskell, and Nix-first infrastructure. Current Bates work spans enterprise authentication, SAML/LTI/Shibboleth, secret management, ACME certificate automation, OpenTelemetry, GitLab AutoDevOps, OpenTofu, RKE2/Rancher, legacy modernization, and 24/7 CVE response. Earlier work shipped Merlin Sound ID and its tooling/demos for fine-grained audio classification used by millions. FOSS and client work spans Apache Solr, rspamd, Postfix, Chapel-lang, GDAL/OGR, Klipper, Marlin, KeePassXC, nixpkgs, xCaddy/libdns, the Svelte/Skeleton ecosystem, ShikiJS, Rocky Linux RPM kernel lanes, Zig capability libraries, GIS, and maker tooling. Comfortable moving between user-facing product surfaces, infrastructure, ML tooling, and low-level debugging.

Always building: Hey! I am *always* hacking, learning, building, reading, and tinkering. Day in, day out, this is what I do. For a more up-to-date view into what I am up to whenever you are reading this document, I invite you to explore my blog at transscendsurvival.org and recent commits on github.com/Jesssullivan.

Product / Applied ML	**Security & Systems**	**Languages & Runtimes**	**Build / CI / Infra**
SvelteKit, React	Static analysis: clang-tidy, scan-build	C / C++ (8+ yrs), Rust, Zig	LLVM+Clang, GCC, Make, CMake
Python/TensorFlow, Flask	Sanitizers: ASan, UBSan	Chapel, Haskell, Futhark	Bazel, autoconf, OpenTofu
Fine-grained classification	Dynamic instr: Frida, Ghidra, ILSpy	Python, Go, R/Rcpp	GitLab CI, GitHub Actions
WASM inference, WebGPU	Fuzzing: AFL++, QuickChpl, cargo-fuzz	Objective-C/Swift, TypeScript	Podman/OCI, Docker, RKE2
OAuth/TOTP/RBAC, SAML/LTI	CVE backporting (kernel, PREEMPT_RT)	Vala (Budgie DE), Bash	Nix flakes, FPM, multi-arch RPM
ML annotation & MLOps tooling	PCIe / NVMe / USB internals	OS internals, ABI / FFI surfaces	OTel, Tempo, SOPS+age, ACME

Selected Systems, Compilers & Applied R&D Work

Compiler & toolchain —

- **Chapel-lang (LLVM-21 backend)** — committer to the Chapel Mason registry and downstream consumer of Chapel's LLVM-based code-generation pipeline. Proficient in Chapel's LLVM 21 lowering, locale-aware codegen, and GASNet `ofi/ucx` substrates. Author of **QuickChpl** — a property-based fuzzing and test harness for Chapel with QuickCheck-style shrinking over distributed code paths.
- **Merlin Sound ID (Clang/LLVM toolchain)** — production iOS audio classifier shipped to millions of users; Objective-C, Swift, Python/TensorFlow training, FFT.js WASM front-end; `libclang`, `dsymutil`, cross-compilation matrices (role detail in §2).

FOSS & Public Production work (committer, contributor, operator) —

- **Klipper** (C, *committer*) — MCU-side firmware and host-side message bus across ARM/AVR HALs, Python↔C interop, and the deterministic-timing real-time event loop; resonance-cancellation tuning and piezo + ADXL345 sensor-fusion algorithm work at the toolhead.
- **KeePassXC** (C++, *committer*) — ongoing security bug fixes and contributions to the macOS build and test coverage.
- **rspamd** (C, C++, Lua, *committer*) — mail-content security at scale; work across rspamd's event loop, libucl config tree, Lua FFI bridge, and the C-level fuzzy/Bayes hash subsystems.
- **Budgie DE** (C, Vala, collaborator) — Vala panel applets for Rocky Linux and ARM64; X11 windowing-stack work for headless and QEMU-based systems.
- **GDAL** (C++) — four years of production work in the GDAL/OGR codebase for GIS and remote-sensing contracts (NPS, NBRC, GPRED): custom drivers, vector/raster pipelines, R bindings via `Rcpp`, KML/CSV/SHP conversion at scale.
- **FFT.js** (Cooley–Tukey FFT implementation, *committer*) — shipped inside Merlin Sound ID; in active use by thousands of ornithologists today.

Kernel & low-level security research —

- **Rocky linux-xr & PREEMPT_RT lane** — carry XR-display patches and a Dirty Frag CVE backport set: **CVE-2026-31431** (algif_aead Copy-Fail), **CVE-2026-43284** (Dirty Frag / xfrm-esp, CVSS 8.8), and **CVE-2026-43500** (Dirty Frag / RxRPC) backported into 6.1.y ahead of public disclosure as C patches against upstream LTS. The `6.1.y-rt` (PREEMPT_RT) sub-lane received additional locking-context audits for `raw_spinlock_t` vs. `spinlock_t` regressions surfaced by the backport set.
- **NVMe XRAM injection paper (2026)** — first-of-its-kind recovery of firmware write-protected NVMe SSDs by injecting NVMe Submission Queue entries into the ASMedia ASM2362 bridge's internal XRAM via vendor SCSI commands, bypassing the bridge's opcode whitelist; end-to-end C, Zig, and inline assembly with PCIe TLP doorbell signaling.
- **Reverse-engineering surface** — `GhidraScript`, `Frida`, `ILSpy`, MITRE Caldera; firmware RE on the hiberpower-ntfs stack; `r2/rizin`, `angr`, AFL++ corpus management, and ASan/UBSan-instrumented harnesses.
- **Zig capability libraries with C ABIs** — author of `zig-crypto`, `zig-notify`, `zig-keychain`, and `zig-ctap2` — C-ABI surfaces with cross-platform builds and hand-written FFI test harnesses.

Rust & modern systems —

- **Rust SIMD** — portable-SIMD and `std::simd` for fine-grained classification kernels feeding WASM inference pipelines.

- **Rust systems work** — in-progress encrypted sync filesystem (Linux FUSE + macOS FileProvider) anchored on `XChaCha20-Poly1305` + `Argon2id`, `FastCDC` content-defined chunking, `BLAKE3` hashing, `zstd` compression, and NATS JetStream-backed fleet sync; Git-safe restore proofing across CLI / daemon / TUI surfaces.

Recent Public Contributions & PRs —

ShikiJS textmate-grammars-themes	svelte-superforms	numtide/nix-vm-test
Futhark	cmux	nixpkgs
xCaddy & libdns	charmbracelet/crush	

Professional Experience

Systems Analyst (DevSecOps) @ Bates College *(2024–Present)*

Build and maintain enterprise systems supporting staff, faculty, and the technical ILS team. 24/7 on-call for CVE triage and mitigation; lead IaC and dev-CI parity college-wide.

- Built a property-based orchestrator and instrumentation harness for 1980s-era degree-management C in `Haskell` + `Python` (QuickCheck PBT, Cabal, FPM-packaged, podman-compose dev parity, GitLab AutoDevOps); turned an unautomatable legacy codebase into a verifiable, traceable, k8s-friendly workload.
- Overhauled and automated the event-management lifecycle (`C#`, `Go`, `Ansible`); led horizontally-scalable `Apache Solr` adoption across multiple public/private indexing applications.
- Built internal ACME-first certificate-management and DNS libraries; led SAML / Shibboleth / LTI integrations; drove enterprise secret management with `KeePassXC` inside a declarative Ansible workflow alongside firewalld, fail2ban, and OTEL / LGTM / Tempo telemetry.

Computer Vision Software Engineer @ Macaulay Library, Cornell Lab of Ornithology *(2019–2021)*

Developed & launched **Merlin Sound ID**, a production fine-grained audio classification system used by millions worldwide. Stack: Objective-C + Swift (iOS), Python/TensorFlow (training), FFT.js (WASM front-end). Led R&D on internal ML annotation tooling, model-evaluation web APIs, and end-to-end MLOps pipelines (Flask, TypeScript, React, Vue, Docker, BitBucket CI); built real-time inference demos in React Native and Swift; coordinated with the Visipedia / iNaturalist community on fine-grained classification benchmarks.

Fabrication Laboratory Manager @ Cornell CALS Landscape Architecture (DLA Makerspace) *(2021–2022)*

Developed and taught rapid-fabrication curricula. Built C++ tiler / slicer tools with my teaching assistants; ran the team's GitHub & Linear project tracking.

Full-Stack Contracting & FOSS (concurrent throughout) *(2017–Present)*

Independent contractor; active upstream contributor and committer (see also §1 for systems depth).

- **Other FOSS / community engagement:** Apache Foundation, Liqo, Joplin, Skeleton UI, ggplot2, and the Rocky Enterprise Linux Foundation. Public PR set called out as bullets in §1.
- **Adaptive Motorsport** (2018): early-stage hardware + systems work.
- **Web GIS** for the National Park Service, Foundation for Healthy Communities, GPRED, NBRC; presented at 2019 AAG Annual Meeting, Washington DC. **Fine-grained classification** R&D with MushroomObserver.org & Visipedia.
- **Independent applied-systems R&D** (2024–present) — agent orchestration for semi-autonomous infrastructure lifecycle management; source-available where appropriate.

Community, Publications & Talks

Community — Ongoing engagement with the **Rocky Enterprise Linux Foundation** (Rocky 10 platform, RPM kernel lanes, and downstream CVE backport practice). *First Fellow*, D&M Makerspace, Plymouth State University (2017–2020); taught *Advanced GIS Programming* and coordinated the regional COVID-19 PPE manufacturing response with the NH hospital system. *Membership Chair & 3D Printing Captain*, Ithaca Generator (2020–2022).

Publications —

Sullivan, J. (2026). *Recovering Write-Protected NVMe SSDs Through USB-Bridge XRAM Injection: Bypassing the ASMedia ASM2362 Firmware Opcode Whitelist.* transscendsurvival.org/papers/recovery-paper.pdf.

Reitsma, L.R., Burns, C., & **Sullivan, J.** (2019). *Poecile atricapillus* (Black-capped Chickadee) Feeding *Catharus guttatus* (Hermit Thrush) Nestlings. *Northeastern Naturalist*, 26(2). doi:10.1656/045.026.0213

Sullivan, J. (2019). *Web GIS: Telling Stories & Solving Problems. Association of American Geographers Annual Meeting*, Washington, DC.

Ten or more professional references (academic, defense-adjacent, FOSS, and industry) available on request.